

KRASNODAR, Russian FederationWMO: **349290**Lat: **45.033N**Lon: **39.150E**Elev: **112**StdP: **14.64**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.6	13.4	1.2	5.9	7.2	7.9	8.3	14.4	26.7	39.4	23.0	32.8	9.3	70	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	94.9	72.5	91.5	71.6	88.3	70.5	75.6	88.2	74.0	86.3	72.5	84.2	10.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	117.9	82.2	70.0	110.9	80.1	68.4	104.6	78.2	39.2	87.7	37.6	86.8	36.3	83.8	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.2	20.6	18.6	-1.4	99.3	11.6	3.5	-9.8	101.8	-16.6	103.9	-23.1	105.9	-31.5	108.4		
			-2.0	78.5	11.5	1.9	-10.2	79.9	-16.9	81.0	-23.4	82.0	-31.7	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	32.5	34.9	43.2	53.7	63.6	71.3	76.1	75.9	66.4	54.4	43.5	35.7
	DBStd	17.27	10.14	9.94	7.40	6.76	6.48	5.31	4.78	5.53	6.62	7.98	7.96	9.23
	HDD50	1955	543	426	231	38	1	0	0	1	47	223	445	
	HDD65	4937	1007	842	676	343	106	10	0	62	337	646	907	
	CDD50	3555	1	4	21	148	424	639	809	804	492	183	27	3
	CDD65	1062	0	0	0	3	63	199	345	340	103	9	0	0
	CDH74	10514	0	0	2	62	600	1766	3548	3549	887	99	1	0
	CDH80	4218	0	0	0	8	144	613	1549	1619	268	17	0	0
Wind	WSAvg	8.3	8.6	9.5	10.1	9.3	7.9	7.6	7.5	7.8	7.7	7.6	7.8	8.1
Precipitation	PrecAvg	27.6	2.4	1.7	1.8	2.3	2.7	3.2	2.2	2.1	1.7	1.8	2.6	3.0
	PrecMax	40.3	6.2	5.4	4.6	7.6	6.2	12.1	6.1	7.0	4.3	3.6	4.8	6.6
	PrecMin	17.2	0.3	0.0	0.0	0.4	0.8	0.2	0.1	0.0	0.0	0.0	0.2	0.3
	PrecStd	4.9	1.5	1.3	1.0	1.5	1.3	2.2	1.7	1.7	1.1	0.9	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	62.5	69.9	80.5	87.7	94.7	98.4	98.9	92.8	82.3	70.2	59.6
		MCWB	47.8	50.6	53.9	61.0	68.7	71.3	72.9	71.7	71.2	65.6	56.7	50.7
	2%	DB	51.4	56.8	64.0	75.1	84.1	89.8	94.8	95.1	86.4	76.6	64.1	54.0
		MCWB	45.3	48.2	51.9	59.1	67.0	71.0	73.4	72.1	68.1	63.0	54.6	48.0
	5%	DB	47.8	51.7	59.1	70.5	80.7	86.3	91.4	91.7	82.7	71.6	59.3	51.0
		MCWB	43.5	45.6	49.6	57.3	65.5	70.1	72.3	71.4	66.7	60.8	53.0	46.4
	10%	DB	44.4	47.8	55.1	66.4	77.0	82.8	88.0	89.1	79.1	67.0	55.2	48.0
		MCWB	41.6	43.2	47.6	55.2	64.1	68.4	70.8	70.5	65.0	58.9	50.3	45.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	52.0	55.4	63.5	71.3	75.6	78.3	77.8	72.9	67.0	58.4	52.0
		MCDB	54.6	59.7	66.0	76.2	83.7	87.4	91.8	91.3	87.5	78.2	66.0	57.7
	2%	WB	46.0	48.9	52.8	60.6	68.7	73.5	76.1	75.4	70.3	64.1	55.9	49.1
		MCDB	49.7	55.8	62.3	71.9	80.6	85.3	88.8	88.5	82.1	73.6	62.1	53.4
	5%	WB	43.4	45.8	50.4	58.4	66.9	71.9	74.4	73.7	68.7	61.8	53.2	46.8
		MCDB	46.6	50.5	57.8	68.6	78.0	82.6	86.4	86.2	79.3	69.8	59.1	50.5
	10%	WB	41.1	43.2	48.0	56.3	65.0	70.1	72.7	72.1	66.8	59.6	50.7	44.5
		MCDB	43.9	47.0	53.9	64.9	74.9	79.9	84.2	84.0	76.3	66.3	54.8	47.3
Mean Daily Temperature Range	5% DB	MDBR	12.1	14.7	16.0	19.5	19.9	19.9	21.1	22.1	20.7	18.3	15.1	12.1
		MCDBR	16.4	21.3	24.4	26.5	25.6	24.5	25.4	25.8	25.9	25.7	23.1	17.6
		MCWBR	11.7	14.0	14.5	13.9	12.2	10.5	10.3	10.6	11.9	14.0	14.8	12.4
	5% WB	MCDBR	15.0	19.5	22.6	23.1	23.2	21.0	22.2	22.6	22.7	23.2	21.1	16.3
MCWBR		11.3	13.6	14.1	12.9	11.8	10.0	10.3	10.3	11.3	13.5	14.3	12.0	
Clear-Sky Solar Irradiance	taub	0.348	0.379	0.422	0.459	0.425	0.430	0.454	0.452	0.417	0.390	0.358	0.347	
	taud	2.279	2.205	2.119	2.061	2.251	2.274	2.220	2.221	2.292	2.327	2.352	2.301	
	Ebn at Noon	234	248	253	254	269	267	259	254	253	243	230	220	
	Edh at Noon	27	35	43	49	43	42	43	42	36	30	25	24	
All-Sky Solar Radiation	RadAvg	395	669	963	1480	1856	2029	2052	1836	1355	863	494	334	
	RadStd	60	84	89	155	194	127	106	122	131	91	65	45	